

Smooth representations of p-adic reductive groups

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More properties of smooth reps of p-adic gps [Bernstein - Zelevinsky]

$F/\mathcal{O}_F$ finite ext'n, $G = G(F)$, G/F conn'd reductive gp.

Let $P \subset G$ (F-pts of) parabolic, Levi decomp. $P = MU$.

Consider two functors smooth reps

parabolic
induction

$$\text{Ind}_P^G : \text{Rep } M \longrightarrow \text{Rep } G$$

$$\pi \longmapsto \text{Ind}_P^G \pi = \left\{ \varphi: G \rightarrow \pi \text{ locally constant,} \right.$$

$$\left. \begin{aligned} \varphi(pg) &= \pi(m) \varphi(g) \\ p &= mu \in P \end{aligned} \right\}$$

$$g \cdot \varphi(g')$$

$$= \varphi(g'g)$$

Jacquet module $J_P : \text{Rep } G \longrightarrow \text{Rep } M$

$$V \longmapsto V_U$$

Prop [BZ 3.13] a) Ind_P^G, J_P are both exact functors

b) [Frobenius reciprocity] $\text{Hom}_M(J_P(V), W) \simeq \text{Hom}_G(V, \text{Ind}_P^G W)$

$$\text{i.e. } J_P \dashv \text{Ind}_P^G$$

c) If $W \in \text{Rep } M$ admissible, $\text{Ind}_P^G W$ also admissible

d) [Jacquet] If $V \in \text{Rep } G$ admissible, $J_P V$ admissible.

Proof. b) ✓

a) Suffices to show for all cpt opens $K \subset G$ that

$$W \rightarrow (\text{Ind}_P^G W)^K \text{ is exact.}$$

not needed, as v.s. but works in p -inverted situation

$$\cong \{ f: P \backslash G/K \rightarrow W: f([g]) \in W^{gKg^{-1} \cap P} \}$$

$$gKg^{-1} \cap M \subset M \text{ cpt open}$$

Exactness of $(\text{Ind}_P^G W)^K$ follows from exactness of $W \rightarrow W^N$ for $N \subset M$ cpt open

For J_P , can write $U = \bigcup_{i \in I} U_i \leftarrow$ cpt open subgps.

For cpt open, $V^{U_i} \simeq V_{U_i}$ for V smooth (reduces to case of finite gps)

$$J_P V \simeq \varinjlim V_{U_i}, \quad \varinjlim \text{ exact, and } V_{U_i} \simeq V^{U_i} \text{ is exact}$$

$\Rightarrow J_P$ is an exact functor.

c) Suffices to show if $K \subset G$ cpt open, W admissible, that $(\text{Ind}_P^G W)^K$ finite rk.

This follows from the fact that $P \backslash G/K$ is finite since P is parabolic.

d) If V is admissible, it suffices to show if $K \subset G$ cpt open, then

$$V^K \rightarrow (V_U)^{M \cap K} \text{ is surjective. Choose } K \text{ s.t. it admits}$$

Iwahori decomposition w.r.t. $P = MU$, i.e. $K = (K \cap U)(K \cap M)(K \cap \bar{U})$

Fact (Bruhat) such subgps are cofinal amongst compact opens

For $GL_n(F)$, can take principal congruence subgp.

↑
opposite
unipotent
radical.

Next goal

Thm If $\pi \in \text{Rep } G$ is irreducible, smooth, then π is admissible.

Crucially uses Harish-Chandra's characterisation of cuspidal reps.

Def. $V \in \text{Rep } G$ is cuspidal if for all $P \subsetneq G$ proper parabolic, $J_P(V) = 0$

Reciprocity

$$\left(\Longleftrightarrow \right) \quad \nexists V \hookrightarrow \text{Ind}_P^G W$$

Thm (Harish-Chandra) $V \in \text{Rep } G$, TFAE

(1) V is cuspidal

(2) the mat. coeffs $\tilde{\zeta} \in \tilde{V}$, $\zeta \in V$, $\psi_{\tilde{\zeta}}, \zeta : G \rightarrow \mathbb{C}$ are cptly supp.

modulo centre

(3) Let $G^o \subset G$ subgp gen'd by all cpt subgps of G , then $V|_{G^o}$ finite.

proof that irred. smooth $\Rightarrow$ admissible (using HC) general facts about G^o :

$G^o \subset G$ open normal subgp, G/G^o is free abelian, finitely gen'd, and

$G/G^o \cong \mathbb{Z}^n$ finite $\left(G = GL_n(F), G^o = \{g \in GL_n(F) : \det g \in \mathcal{O}_F^\times\} \right)$

$$G/G^o \cong \mathbb{Z}^n \cong F^\times / \mathcal{O}_F^\times$$

First assume V is cuspidal, HC $\rightarrow V|_{G^o}$ has cptly supp. mat. coeff.

Since $G/G^o \subset G$ is finite index, $V|_{G/G^o} \cong \bigoplus_{i=1}^k V_i \hookrightarrow \text{irred. } G/G^o\text{-mod}$

Schur's lemma $\Rightarrow \mathbb{Z}$ acts as scalar on each V_i

$\Rightarrow V_i$ is irred as a G^o -rep.

previous lecture: irred. + cply supp mat. coelt. $\Rightarrow V_i^*$ admissible.
 $\Rightarrow V$ admissible

If V not cuspidal, choose $P \subset G$ parabolic $\leadsto J_P(V)$ cuspidal as an M -rep'n $\neq 0$

Take $\rho \subset J_P(V)$ irred quot.

$$\text{Hom}_M(J_P(V), \rho) \cong \text{Hom}_G(V, \text{Ind}_P^G \rho)$$

$$V \text{ irred} \Rightarrow V \hookrightarrow \text{Ind}_P^G \rho$$

$\Rightarrow V$ admissible. admissible by first case + previous thm.

Coelt. ring: uses Schur's lemma + $|G/G^0\mathbb{Z}|$ coprime to char. + Harish-Chandra's thm.

(Sketch) show conditions equiv. to: if $\xi \in V$, $K \subset G$, ^{cpt open}

(*_{cuspidal}) $K_\xi = \{g \in G: \pi(g) \pi(g^{-1}) \xi \neq 0\}$ is cpt mod centre

(*_{cuspidal}) $\Rightarrow \forall \xi, \tilde{\xi}$ cpt supp. / $\mathbb{Z}$: $\tilde{\xi} \in \tilde{V}^K$, $\xi \in V$, have

$$\text{Supp } \varphi_{\xi, \tilde{\xi}} \subset K_\xi \leftarrow \text{cpt} / \mathbb{Z}.$$

(2) $\Rightarrow$ (3)

G v.s. G^0

Cpt supp

follows from fact that $\tilde{V}$ indep. of using G or G^0 .

(3) $\Rightarrow$ (*_{cuspidal}) uses [BZ 2.40]

(1) $\Leftrightarrow$ (*_{cuspidal}) uses [BZ 3.22]

Lemma $\xi \in V$, TFAE:

(state for $GL_n(F)$)

(a) $\forall K \subset G$ sufficiently small cpt opens, $\exists t = t(K, \xi)$ s.t.

$$\pi(\xi_K) \pi(s^{-1}) \xi = 0 \quad \text{for all } s = \begin{pmatrix} pa_1 & & \\ & \ddots & \\ & & pa_n \end{pmatrix} \in Z(M)$$

↑
has blocks

mults
= blocks
of M
 $a_i \geq a_{i+1}$,
 $a_1 - a_n > t$.

(b) $\xi \in \ker(V \rightarrow V_K)$

→

Thm (Howe) $V \in \text{Rep } G$ is finite length $\Leftrightarrow$ finitely generated + admissible

proof sketch: $\Rightarrow$: previous result.

$\Leftarrow$: uses that if $K \subset G$ cpt open sufficiently small (for $GL_n(F)$)

then if a finite set $a_1, \dots, a_n \in V^K$ generates V , then

any G -subspace $W \subset V$ is gen. by W^K .

(see congruence subgrp)

using fact if $V_0 \subsetneq V_1 \subsetneq V_2 \subsetneq \dots \subset V_k = V$

then passing to K -ints preserves nontrivial pieces

$$V_0^K \subsetneq V_1^K \subsetneq V_2^K \subsetneq \dots \subseteq V_k^K = V^K$$

V^K finite rk by admissibility, so flag has length $\leq \text{rk } V^K$. $\square$

(FACT) distinguish b/w cuspidal + noncuspidal cases in proof.

try to show irred cusp sub is (roughly) a direct summand.
uses Harish-Chandra

irred smooth $\Rightarrow$ adm. tells us for any $K \subset G$ cpt open that any simple K_K -module is finite rank.

Thm If $K \subset G$ cpt open. $\exists C = C(G, K)$ s.t. any irred. K_K -mod has rank $\leq C$.

Proof sketch uses a linear alg. lemma.

if V has $\dim = n$, $R \subset \text{End } V$ comm. subalg gen'd by $a_1, \dots, a_\ell, 1$,

$$\text{then } \dim R \leq n^{2-2^{1-\ell}}$$

Apply lemma to K_K -module str. map $\rho: K_K \rightarrow \text{End } V$

General algebra (Burnside) $\rightarrow \rho(K_K) = \text{End } V$.
 (irred. f.d. by irr. sm $\rightarrow$ adm.)

Passing to $A \subset K_K$ comm. subalg gen'd by

$\Lambda = \frac{X_*(T)}{\text{max. spl. + tors of rank } \ell}$ free abelian gp. then

$$\dim \rho(A) \leq (\dim V)^{2-2^{1-\ell}}$$

General fact, if $C = [K \setminus \Gamma]$, then $K_K = \sum_{i,j=1}^C \delta_{ij} + A + \delta_j$
 $\uparrow$ max. cpt $\uparrow$ coset reps of $K \setminus \Gamma$

$$(\dim V)^2 \leq C^2 (\dim V)^{2-2^{1-\ell}} \Rightarrow \dim V \leq C^{2^\ell}$$

Use this fact to prove:

Thm $\bigcup_{\text{Rep } G} \text{finitely gen'd}$ is noetherian

$\Rightarrow K_K$, $K \subset G$ cpt open finitely generated + noetherian.